Arnis 3.4.0 and Confluence.4

What differs, and what it costs on a large area. Confluence.4 is a test build from the fork, not an official Arnis release.

1. Features

Feature	Arnis 3.4.0	Confluence.4
One World mode (many regions, one world folder)	yes	yes
Build in pieces with parallel workers (`--unit-regions`, `--one-world-workers`)	no	yes
Auto piece size and worker count from RAM and CPU	no	yes
RAM budget and thread limit (`--ram-budget-mb`, `--threads`)	no	yes
Resume a job, rebuild chosen pieces (`--skip-built`, `--only-pieces`)	no	yes
Piece plan on the map, with time and RAM estimate	no	yes
Map legend (? button)	no	yes
Projects tab: open a world, expand it	no	yes
Download now (fill the caches, build nothing)	no	yes
Offline mode, local tile archive, Region Download	no	yes
Water detail, river bed, climate mode, seed, height limits	no	yes
Rocks, bushes, tree packs, props	no	yes
Cave style and cave ores	no	yes
Road detail, no-buildings, chest loot table	no	yes
B_Linear region format, world border, origin	no	yes
Polygon Area shapes for selecting land	no	yes (from upstream Arnis)
Multi-area merge	no	yes (from upstream Arnis)
Discord status	no	yes (from upstream Arnis)
Parallel path for Bedrock output	no	yes (from upstream Arnis)
Faster Java NBT writer	no	yes (from upstream Arnis)

With Advanced features off, the world matches stock Arnis main on the test area (0 of 840 files differ). Thirteen window controls from the 3.4.0 fork are not back; their flags still work on the command line.

2. Speed and RAM on three areas

Stock 3.4.0 against the tuned Auto setting (`--one-world --unit-regions auto --one-world-workers auto`). Ratio = tuned Auto against stock; a minus sign means tuned Auto is lower. No run timed out or ran out of memory.

Area	Wall time, median of 3			Peak RAM			World size	
	Stock	Tuned Auto	Ratio	Stock	Tuned Auto	Ratio	Stock	Tuned Auto
S8 (64 km²)	15.3 s	16.3 s	+6.5%	2.5 GB	2.5 GB	+0.4%	1.0 GB	1.0 GB
S16 (256 km²)	57.9 s	52.8 s	-8.8%	7.2 GB	3.4 GB	-53.1%	3.8 GB	3.9 GB
S24 (576 km²)	94.4 s	103.1 s	+9.2%	7.9 GB	4.8 GB	-39.0%	8.6 GB	8.8 GB

Pieces trade a little speed for much lower RAM. Stock is faster at 64 km² (15.3 s against 16.3 s; tuned Auto builds this size as one run) and at 576 km² (94.4 s against 103.1 s). At 256 km² tuned Auto took 52.8 s against 57.9 s, with 3.4 GB peak RAM against 7.2 GB (-53.1%); at 576 km² the RAM is 4.8 GB against 7.9 GB (-39.0%). Pieces write a slightly larger world (+2.4% at 256 km²).

Sweep: piece size and worker count

Regions per piece side / workers	2 workers	4 workers	6 workers
2×2	140.3 s (137.9–141.1) 0.7 GB	101.9 s (100.8–103.0) 1.1 GB	94.8 s (94.7–146.3) 1.4 GB
4×4	83.3 s (82.4–85.6) 1.3 GB	68.2 s (66.6–71.3) 2.1 GB	71.0 s (70.1–72.2) 2.6 GB
8×8	71.1 s (67.8–72.1) 2.5 GB	64.9 s (64.8–67.2) 3.2 GB **Recommended**	72.0 s (69.7–74.0) 4.3 GB

Sweep on 256 km² (S16), median of 3 (min–max) and peak RAM, run before the tune. The best median is 64.9 s (8×8 pieces, 4 workers); the next, 4×4 with 4 workers, is 68.2 s, outside the best cell's range, so the tie rule does not apply and the fastest cell wins. Tuned Auto matches the best sweep result only at 256 km²: 52.8 s against 52.8 s for the fixed cell, both in the second session.

These results come from one dense city on one machine. A rural area, a slower disk or less RAM can move them.

Machine: 24 threads, 31 GB RAM, Windows 11; one machine, three areas. Runs on 2026-11-10 and 2026-11-11. Stock = Arnis 3.4.0 exe (sha256 aafa66e9…). Confluence.4 = pre-tuning gate build, commit 62507023 (sweep), plus the Confluence.4 tuned build (arnis-3.5 at f44c4678, tuned Auto 1cefd515, sha256 b295e96b…) for the tuned numbers. Caches warm; the two cold-cache runs (one each) are in docs/23-BENCH-PIECES-VS-SINGLE.md. Wall time is the median of 3 runs; the grid also shows min–max. The second session ran about 20% faster than the first (stock S16 72.7 s, then 57.9 s), so the tuned comparison uses stock reruns from the same session. Failed, timed-out or out-of-memory runs: 0. Source: docs/bench/results*.jsonl, built by make_pdf.py.